RECOVERY FINE-TUNING RECOVERS WHERE IT WAS TRAINED: DURATION- AND DOMAIN-DEPENDENT GAINS IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Structured pruning of text-to-audio diffusion models is followed by recovery fine-tuning, whose benefit is usually certified by one score at one inference setting. Using the released pruned and fine-tuned AudioLDM-M checkpoints at two pruning severities (65% and 83% of U-Net parameters removed), we measure the paired recovery gain in text–audio alignment across clip duration and prompt domain, anchored by a shuffled-caption chance floor and the real audio of the same prompts. At 83% pruning, fine-tuning closes 63% of the pruned checkpoint's gap to real audio at its own 10.24 s fine-tuning duration but only 33% at 3.84 s, and nothing on held-out hip-hop captions; against the unpruned model, 82% versus 44% at 83% and 52% versus 8% at 65%. The duration dependence was pre-specified, replicates on a disjoint prompt set, survives a family-wise correction, is not a scorer artefact, and is reproduced at both durations by a second scorer and by two event-level metrics outside the CLAP family. Recovery should be reported across operating points; lacking a matched dense fine-tuned control and human ratings, our claims concern evaluation, not mechanism.

Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is the standard route to cheaper diffusion models [1–3], recently applied to text-to-audio (TTA) generation with AudioLDM [4, 5]. Recovery is judged by the restoration of one aggregate number on the in-domain test set at one sampler setting. A generative model, however, is used at an inference operating point: a clip duration, a prompt domain, a sampler configuration. Recovery fine-tuning is itself training at one operating point; nothing guarantees that a gain measured there holds elsewhere.

We ask: does the gain of recovery fine-tuning over the pruned checkpoint depend on the operating point at which it is evaluated? We take the released pruned and fine-tuned AudioLDM-M-Full checkpoints of Singh et al. [5] at two pruning severities, train nothing, and measure the paired recovery gain in text–audio alignment across two clip durations and two prompt domains, read against a chance floor and a real-audio ceiling. Our findings:

- Duration. The recovery gain is several times larger at the fine-tuning duration (10.24 s) than at 3.84 s, at both severities. The interaction was pre-specified, replicates on a disjoint prompt set, and is not a scale or scorer effect (the chance floor moves by at most 0.025 between durations).
- Domain. At matched duration the gain is present in-domain (AudioCaps) and absent or negative on held-out hip-hop/rap captions—at both severities at 3.84 s and at both durations at severity 2—where alignment after fine-tuning sits 0.02–0.03 above the chance floor, against 0.32 in-domain.
- Where the gain lives. A frame-level grounding model shows the native-duration gain spread uniformly over the clip; a crop analysis shows the short-duration deficit arises from generating short clips, not from scoring short excerpts.

2. BACKGROUND

Pruning and recovery in diffusion models. Structural pruning with fine-tuning recovery is established for image diffusion models [1–3] and self-supervised speech encoders [6, 7]. For TTA, Singh et al. [5] L1-prune the AudioLDM U-Net, fine-tune on AudioCaps for 10^6 steps and report in-domain FAD/KL at a fixed sampler; their fine-tuned pruned model even surpasses the unpruned one in-domain (FAD 1.57 vs. 3.95), a first sign that recovery also acts as in-domain specialisation. All evaluate recovery at one operating point.

Evaluating text-to-audio generation. CLAP similarity [8] is the standard automatic proxy for text–audio alignment [9,10]; FAD [11] is sensitive to sample size and embedding [12, 13]; event classifiers [14], human-aligned CLAP [15] and frame-level grounding [16] measure other axes, motivating a paired, multi-metric design with one pre-specified primary endpoint.

Duration as an operating point. Latent audio diffusion conditions on the latent length and is trained at a fixed duration (10.24 s for AudioLDM [4, 17]; newer systems condition on timing explicitly [18]), so duration is mechanistically motivated for a checkpoint fine-tuned at one duration; fine-tuning also trades in-distribution gains for out-of-distribution losses [19], motivating the domain axis.

3. EXPERIMENTAL DESIGN

3.1. Systems

The dense model is AudioLDM-M-Full [4] (U-Net stage multipliers $(1, 2, 3, 5)$, 415.96 M U-Net parameters), itself AudioCaps-fine-tuned by its authors. We study the two released pruning severities of [5]: severity 1, $(1, 2, 3, 1)$, 145.67 M parameters (-65.0%), and severity 2, $(1, 2, 1, 1)$, 71.08 M (-82.9%; 39% fewer multiply–accumulates [5]). At each severity we compare the pruned checkpoint (P) with the fine-tuned checkpoint (P+FT), the released result of 10^6 steps of AudioCaps recovery fine-tuning of P [5] (EMA weights, checksums verified). The released pruned checkpoints carry no usable EMA weights, so P is the released L1 channel selection applied to the dense EMA weights (bit-exact to the released raw weights at severity 1; at severity 2 the release differs in three decoder-seam tensors, so both conventions are evaluated: A′ primary, B′ sensitivity). "Recovery" names the fine-tuning stage, not an achieved restoration; we train no system. Two of the present authors released the evaluated checkpoints [5]; the operating points, batteries, estimands and gates below were specified independently of that work.

3.2. Operating points and prompt batteries

Two durations: the native point (10.24 s, latent length 256; the fine-tuning duration) and a short point (3.84 s, latent length 96). Two domains: in-domain AudioCaps test captions [20] and held-out hip-hop/rap captions from MusicCaps [21], a sub-domain with near-zero exposure in the recovery corpus ("music"; its captions are seven times longer, median 56.5 vs. 8 words, so the domain axis bundles content with caption style). Severity 2 uses 192 AudioCaps prompts (disjoint from all severity-1 prompts) and 64 music prompts (3 replicates at 3.84 s, one at 10.24 s); severity 1 uses 96 AudioCaps prompts for the pre-specified domain test, a matched 80-prompt subset for the duration comparison, and 64 music prompts at 3.84 s; the dense model is generated at both durations on both AudioCaps sets (matched controls, same noise). Generation uses DDIM [22], 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights (off the published recipe, Sec. 5).

3.3. Paired protocol, scoring and anchors

Within each operating point the compared systems receive identical generation noise x_T per prompt (common random numbers); across durations the latent shapes differ, so duration contrasts are prompt-paired but not noise-paired. The primary scorer is fused CLAP (laion/clap-htsat-fused, rev. 365dea6e) [8], which repeat-pads 3.84 s audio to 10 s and centre-crops 10.24 s audio to 10 s; every (system, operating point) cell is scored with one fixed-order, fixed-seed call, part of the endpoint. Two anchors make absolute scores readable: a cell's chance floor, the mean cosine between each clip and the captions of the other prompts in its battery (a shuffled-caption baseline from the same embeddings, so the matched scores are unchanged), and the real-audio ceiling, the real AudioCaps clip of each prompt, band-limited to 16 kHz and scored under the identical convention at full length and as its first 3.84 s. The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap intervals ($B = 10^4$).

3.4. Quantities and analysis plan

Let R be the mean per-prompt recovery gain, CLAP(P+FT) − CLAP(P), at a given (domain, duration); R_{nat}, R_{short}, R_{mus} denote AudioCaps-native, AudioCaps-short and music-short. The duration interaction is $J = R_{\mathrm{nat}} - R_{\mathrm{short}} = s(P{+}FT) - s(\mathrm{P})$, where the duration response $s(\cdot)$ is a system's mean per-prompt score change from 3.84 to 10.24 s; J is paired per prompt, with pre-specified gate $\mathrm{lo}_{95}(J) > 0$. The recovery ratio $\rho = R/(\mathrm{ref} - \mathrm{P})$ is the fraction of the pruned checkpoint's gap to a reference (dense, or the real audio of the same prompts) that fine-tuning closes. The domain contrast at matched duration is $R_{\mathrm{short}} - R_{\mathrm{mus}}$; the smallest effect size of interest is 0.025. "Pre-specified" means that estimand, gate and prompt set were committed to the version-controlled repository before any score was seen: the severity-1 domain test (the original reversal hypothesis; failed), the severity-1 duration follow-up, the severity-2 replication with seam sensitivity (primary), the severity-2 music cell at 10.24 s and the secondary metrics. The FineLAP diagnostic and the severity-2 dense control were registered after the primary result: committed before their own scores were seen, after the primary endpoint had been read. The severity-1 dense control, crop, rank-scale, Holm and anchor analyses are post-hoc. These three labels carry no other meaning below.

4. RESULTS

4.1. The recovery gain grows with clip duration

At the more severe operating point, recovery fine-tuning buys several times more alignment at its own duration than at 3.84 s. On the disjoint 192-prompt set, with estimands frozen before scoring, the interaction is clearly resolved, $J = +0.159$ (Fig. 1, Table 1). Fine-tuning raises CLAP alignment from 0.055 to 0.299 at the native duration ($R_{\mathrm{nat}} = +0.244$) but only from 0.015 to 0.100 at 3.84 s ($R_{\mathrm{short}} = +0.085$; resolved, not absent); the pruned checkpoint gains just $s(\mathrm{P}) = +0.040$ from the longer clip, the fine-tuned one $s(P{+}FT) = +0.200$. The seam convention does not matter (B′: $J = +0.161$).

The effect is not an artefact of scale. The chance floor is close to zero and nearly duration-independent (Table 2: -0.048 to $+0.020$ across the AudioCaps cells, moving by at most 0.025 between durations for any system), so the interaction survives chance correction: $J_c = +0.191$ [+0.162, +0.220]. Real audio of the same prompts scores 0.274 at 3.84 s and 0.440 at 10.24 s ($s(\mathrm{real}) = +0.167$ [+0.150, +0.184], the scorer's own response to seeing the whole clip). Against that ceiling, fine-tuning closes $\rho = 63\%$ of the pruned checkpoint's gap at the native duration but 33% at 3.84 s.

The matched dense control separates the systems' duration response from the scorer's. At severity 1, on the same 80 prompts and under the same convention, the dense model's response $s(\mathrm{dense}) = +0.150$ ($0.202 \to 0.352$) equals the pruned checkpoint's, $s(\mathrm{P}) = +0.149$ (paired difference -0.001 [-0.056, +0.055]), while the fine-tuned checkpoint responds more, $s(P{+}FT) = +0.193$ (excess over dense +0.043 [-0.020, +0.109]). The recovery gain itself is small there ($R_{\mathrm{short}} = +0.008$, $R_{\mathrm{nat}} = +0.052$, $J = +0.044$), and part of

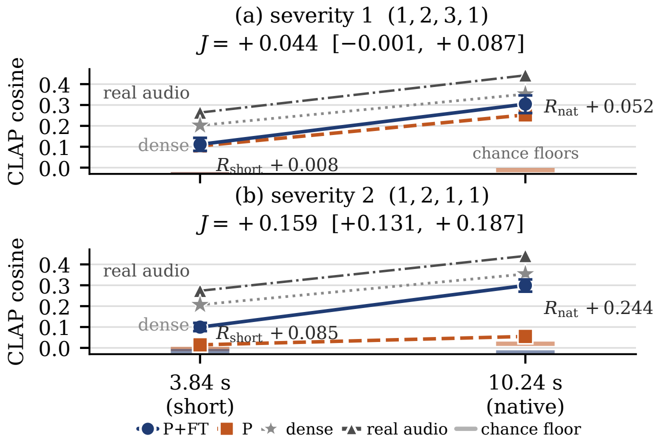

Fig. 1. The recovery gain grows with clip duration. Mean CLAP cosine of the pruned (P, dashed) and fine-tuned (P+FT, solid) checkpoints at the short and native points, (a) severity 1, (b) severity 2; whiskers: 95% CI of the paired gain R about the P+FT mean. Anchors: real audio of the same prompts (triangles), each cell's chance floor (ticks) and the matched dense control (stars; 80 prompts at severity 1, 192 at severity 2, same scoring convention).

the pruned checkpoint's raw response is a rise of its own floor (+0.019; the fine-tuned checkpoint's floor does not move), so on chance-corrected scores the severity-1 interaction is resolved, $J_c = +0.066$ [+0.020, +0.111]. In recovery ratios, fine-tuning closes 8% of the pruned checkpoint's gap to dense at 3.84 s and 52% at 10.24 s (Table 2). The severity-2 control reads the same on its own prompts ($0.207 \to 0.354$): $s(\text{dense}) = +0.147$ [+0.117, +0.177], against which the pruned checkpoint responds −0.107 [−0.137, −0.076] less and the fine-tuned checkpoint +0.053 [+0.017, +0.089] more (paired); fine-tuning closes 44% of the gap to dense at 3.84 s and 82% at 10.24 s (Table 2).

Severity 1 is underpowered, not negative: its raw-scale interval narrowly includes zero, but the follow-up was powered only for $J \geq 0.065$ at $n = 80$. It is directionally consistent on every scale (post-hoc): W rises from 0.44 to 0.64 ($\Delta W = +0.20$ [+0.06, +0.34]), the median per-prompt interaction is +0.051 [+0.012, +0.077] and the second scorer gives $J_{\text{HC}} = +0.075$ [+0.012, +0.137]; at severity 2, $\Delta W = +0.15$ [+0.07, +0.22] and the median interaction is +0.172. A Holm correction over all 13 contrasts leaves every severity-2 conclusion unchanged ($p < 10^{-4}$); at severity 1, R_{nat} ($p = 0.016$) and J ($p = 0.052$) do not survive, so the severity-1 duration effect rests on the severity-2 replication, not on its own.

4.2. At matched duration the gain is domain-specific

The gain does not transfer off the fine-tuning domain, at either duration. At severity 2 and 3.84 s it is resolved in-domain ($R_{\text{short}} = +0.085$) and null on music ($R_{\text{mus}} = +0.009$; $W = 0.48$), a matched-duration domain contrast of +0.076; at the native duration the music contrast is likewise null (+0.005; $W = 0.53$), so the large native gain is confined to the fine-tuning domain (domain contrast +0.239; music duration interaction −0.004 [-0.043, +0.034]). At severity 1 the pre-specified domain test (96 prompts) found no in-domain gain at 3.84 s ($R_{\text{AC}} = -0.002$ [−0.027, +0.021]) and a negative music contrast ($R_{\text{mus}} = -0.094$; domain contrast +0.092).

The chance floor reads the two severities as one statement (Table 2): after fine-tuning, alignment on the hip-hop captions is 0.022, 0.018 and 0.033 above chance (severity 1 at 3.84 s; severity 2 at 3.84 and 10.24 s), against 0.115 and 0.321 in-domain at severity 2. Whether that registers as a "penalty" depends only on whether the pruned checkpoint still had music alignment to lose: the 65%-pruned one did (0.061 above chance; chance-corrected $R_{\text{mus}} = -0.040$ [−0.060, −0.020]), the 83%-pruned one did not (0.018). The null is a null near chance for both checkpoints, as CLAP measures it (no real hip-hop reference was available for a ceiling), and not a caption-length effect: within AudioCaps the native gain is uncorrelated with caption length (Spearman $\rho = +0.04$ [−0.12, +0.18]), and no caption is truncated by the conditioner—though 47% of the music captions exceed CLAP's 77-token pre-training length, a caption-style covariate inseparable from content.

Table 1. Recovery gain in CLAP alignment: mean cosine per checkpoint, paired gain $R = \text{CLAP}(P{+}FT) - \text{CLAP}(\text{P})$ with 95% CI, and fraction W of prompts on which P+FT beats P. Duration rows give $s(\text{P})$, $s(P{+}FT)$ and, in the R column, J (paired per prompt; domain contrasts two-sample; n as in Sec. 3.2). †: CI excludes 0; ‡: pre-specified gate passes. Holm survivors: all severity-2 †, severity-1 music rows.

Setting	P	P+FT	R [95% CI]	W
Severity 1, (1, 2, 3, 1)				
AudioCaps 3.84 s	0.104	0.111	+0.008 [-0.023, +0.039]	0.44
AudioCaps 10.24 s	0.253	0.304	+0.052† [+0.009, +0.093]	0.64
music 3.84 s	0.117	0.023	−0.094† [-0.124, -0.065]	0.20
duration s; J	+0.149	+0.193	+0.044 [-0.001, +0.087]	
domain 3.84 s (n=96)			+0.092† [+0.054, +0.131]	
Severity 2, (1, 2, 1, 1)				
AudioCaps 3.84 s	0.015	0.100	+0.085† [+0.066, +0.105]	0.72
AudioCaps 10.24 s	0.055	0.299	+0.244† [+0.215, +0.273]	0.87
music 3.84 s	0.005	0.014	+0.009 [-0.013, +0.032]	0.48
music 10.24 s	0.089	0.094	+0.005 [-0.028, +0.039]	0.53
duration s; J	+0.040	+0.200	+0.159‡ [+0.131, +0.187]	
domain 3.84 s			+0.076† [+0.047, +0.105]	
domain 10.24 s			+0.239† [+0.195, +0.283]	

Table 2. Anchors and recovery ratios (AC = AudioCaps; $n = 80$ at severity 1, 192 at severity 2, 64 music): chance floor of the P/P+FT cells, mean CLAP of the real audio of the same prompts, and the fraction ρ of the pruned checkpoint's gap to dense and to real audio closed by fine-tuning (95% CI, in %). Dense control: severity 2 registered after the primary result, severity 1 post-hoc (Sec. 3.4).

Setting	floor P/P+FT	real	ρ_{dense} [%]	ρ_{real} [%]
sev. 1 AC 3.84 s	−0.031/ − 0.033	0.264	8% [−30, 36]	5% [−16, 24]
sev. 1 AC 10.24 s	−0.012/ − 0.036	0.442	52% [11, 103]	27% [6, 46]
sev. 2 AC 3.84 s	−0.005/ − 0.015	0.274	44% [36, 53]	33% [26, 39]
sev. 2 AC 10.24 s	+0.020/ − 0.022	0.440	82% [72, 92]	63% [56, 71]
sev. 1 music 3.84 s	+0.055/ + 0.001	–	–	–
sev. 2 music 3.84 s	−0.013/ − 0.004	–	–	–
sev. 2 music 10.24 s	+0.070/ + 0.061	–	–	–

4.3. Where in the clip the gain lives

Two analyses locate the native-duration gain. First, as a prospectively frozen diagnostic, we score the native clips

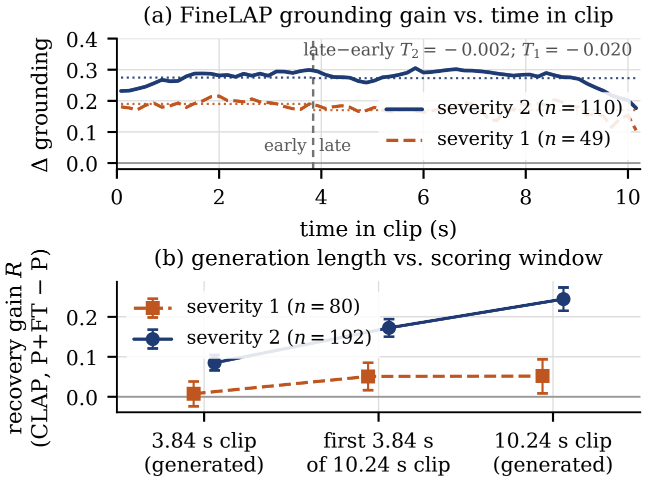

Fig. 2. (a) FineLAP frame-level grounding gain (P+FT − P) vs. time in the 10.24 s clip (dotted: window means; vertical line: 3.84 s): uniform, not back-loaded. (b) Recovery gain R (95% CI) on the generated 3.84 s clip, the first 3.84 s of the 10.24 s clip and the full clip: a generation-length, not a scoring-window, effect.

with FineLAP [16], a frame-level language–audio grounding model (EAT encoder; not CLAP-derived) reporting per 0.16 s frame how strongly the requested event is grounded (110 and 49 eligible prompts at severities 2 and 1, selected outcome-blind). Fine-tuning raises grounding across the whole clip (severity 2: semantic mass +0.27 [+0.22, +0.33]; both severities; seam-robust), and the gain in the first 3.84 s equals the gain in the remainder: $D_{\text{early}} = +0.275$ vs. $D_{\text{late}} = +0.273$, $T = -0.002$ [−0.024, +0.020] (Fig. 2a). Second, a post-hoc crop analysis separates generation length from scoring window: we score the first 3.84 s of each 10.24 s generation with the same CLAP convention and compare with the separately generated 3.84 s clips of the same prompts (Fig. 2b). The crop carries a large part of the native gain: at severity 1, $R_{\text{crop}} = +0.051$ [+0.016, +0.085] equals R_{nat} ($R_{\text{nat}} - R_{\text{crop}} = +0.001$ [−0.027, +0.029]) and exceeds R_{short} by +0.043 [+0.002, +0.085]; at severity 2, $R_{\text{crop}} = +0.172$ [+0.150, +0.194], $R_{\text{crop}} - R_{\text{short}} = +0.087$ [+0.065, +0.110], $R_{\text{nat}} - R_{\text{crop}} = +0.072$ (seam-robust). The short-duration deficit is thus mostly a property of what the fine-tuned checkpoint generates, not of scoring a short excerpt.

4.4. Corroboration and pre-specified negatives

At the severity-2 native point Human-CLAP [15] (a CLAP model fine-tuned on human ratings; automatic) gives $R_{\text{nat}} = +0.375$ [+0.340, +0.408] and $J = +0.185$; two pre-specified event-level metrics favour P+FT with paired CIs excluding zero: KL to real references, 2.23 vs. 4.45 ($\Delta = +2.22$ [+1.93, +2.53]); PANNs top-10 captured labels, 1.46 vs. 0.60 ($\Delta = +0.86$ [+0.70, +1.02]). FAD agrees (6.92 vs. 27.4, descriptive). Rescored at 3.84 s against the same references cropped alike (post-hoc), both event-level gains shrink—KL +0.66 [+0.42, +0.92], PANNs +0.19 [+0.06, +0.32]—so the interaction holds outside the CLAP family: $J_{\text{KL}} = +1.56$ [+1.19, +1.92], $J_{\text{PANN}} = +0.67$ [+0.49, +0.86] (seam-robust).

Three pre-specified hypotheses failed and are kept: (i) that recovery fine-tuning trades in-domain for out-of-domain alignment—it requires an in-domain gain at 3.84 s, and severity 1 has none ($R_{\text{AC}} \approx 0$); the loss on music is real, the trade is not; (ii) the severity-1 music penalty did not replicate (Human-CLAP alone shows −0.037 [−0.068, −0.005] at severity 2; the sign is unestablished); (iii) a "late allocation" account of the duration effect is rejected by FineLAP ($T \approx 0$). Finally, P+FT is not restored to dense: at severity 1 and 10.24 s the dense model leads P by +0.099 [+0.058, +0.140] and P+FT by +0.048 [−0.000, +0.096]; at severity 2 by +0.055 [+0.021, +0.088] at 10.24 s (+0.107 at 3.84 s).

5. DISCUSSION AND LIMITATIONS

Across severities, scorers and metrics, the recovery gain of the released fine-tuned AudioLDM-M checkpoints tracks the operating point of the fine-tuning itself: largest at the fine-tuning duration, confined to the fine-tuning domain, heterogeneous across prompts (column W). At 65% pruning the pruned checkpoint still responds to clip duration like the dense model, and fine-tuning adds a modest, in-domain-only increment. At 83% pruning it is barely above chance at either duration (0.020 and 0.035 above the floor), and fine-tuning restores a duration response above the dense model's—only in the fine-tuning domain and only at the fine-tuning duration (Table 2), consistent with the recovery stage acting, at least in part, as further in-domain, native-duration specialisation. Post-pruning recovery is thus better described by a few operating-point-specific recovery ratios than by one restored score. Recovery studies should report the training-matched operating point and one motivated off-native point; absolute scores of P, P+FT and dense with a chance floor at each; paired gains with prompt-level uncertainty under common random numbers; and an explicit interaction across points.

Limitations. One case study (one model family, two severities, two durations, two domains). Mechanistic attribution is blocked: the matched control—the dense model given the same AudioCaps fine-tune—no longer exists, and the public dense text-fine-tuned release is not equivalent (different data and recipe), so the dependence cannot be attributed to pruning rather than to fine-tuning in general. Primary inference rests on CLAP-family scorers; there is no human evaluation. Sampler settings differ from the published recipe (DDIM 50 vs. 200, guidance 2.5 vs. 3.5, single vs. best-of-3; both systems evaluated identically). One held-out sub-genre (no real hip-hop references for a ceiling); cross-severity comparisons are descriptive; the severity-1 duration test was underpowered and does not survive family-wise correction. Analysis statuses are as declared in Sec. 3.4.

6. CONCLUSION

Recovery fine-tuning of the released pruned AudioLDM-M checkpoints recovers much at its own operating point and much less away from it: at 83% pruning it closes 63% of the gap to real audio at the fine-tuning duration, 33% at 3.84 s and none on held-out music. One restored score at one operating point does not describe that. Audio: gbibbo.github.io/audioldm-modality-swap-pruning.

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